



Visual–haptic feedback interaction in automotive touchscreens

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ABSTRACT

Touchscreen interfaces offer benefits in terms of flexibility and ease of interaction and as such their use has increased rapidly in a range of devices, from mobile phones to in-car technology. However, traditional touchscreens impose an inevitable visual workload demand that has implications for safety, especially in automotive use. Recent developments in touchscreen technology have enabled feedback to be provided via the haptic channel. A study was conducted to investigate the effects of visual and haptic touchscreen feedback on visual workload, task performance and subjective response using a medium-fidelity driving simulator. Thirty-six experienced drivers performed touchscreen ‘search and select’ tasks while engaged in a motorway driving task. The study utilised a 3×2 within-subjects design, with three levels of visual feedback: ‘immediate’, ‘delayed’, ‘none’; and two levels of haptic feedback: ‘visual only’, ‘visual + haptic’. Results showed that visual workload was increased when visual feedback was delayed or absent; however, introducing haptic feedback counteracted this effect, with no increases observed in glance time and count. Task completion time was also reduced when haptic feedback was enabled, while driving performance showed no effect due to feedback type. Subjective responses indicated that haptic feedback improved the user experience and reduced perceived task difficulty.

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1. Introduction

Touchscreen interfaces are employed across a wide range of technologies, from mobile phones to in-car infotainment systems. Growth in these markets has been rapid: the number of users of touchscreen phones increased by 159% in the year August 2008–2009 [1], with a corresponding increase of 49% in worldwide smartphone sales between Q1 2009 and Q1 2010 [2]. Similarly, the availability of touchscreen-based in-car systems in the US market increased from 44% to 55% for the 2009 model year [3]. These trends reflect the increased functionality present in today’s technology and the flexibility afforded by a touchscreen interface.

Touchscreens offer practical benefits in use [4]: input is direct as the display and input are co-located and the simple mode of interaction, pointing, is familiar even to novice users [5]. A major drawback with ‘traditional’ touchscreens however is the lack of haptic feedback, both tactile and kinaesthetic, provided by the screen’s surface. This places significant demand on visual attention [6] which raises potential issues with respect to safety; road acci-

dent study data indicate that 60% of crashes, near-crashes and incidents can be attributed to glances away from the forward roadway [7].

The provision of feedback is an essential part of the interaction experience. Norman [8] describes the ‘Gulf of Evaluation’ – the effort that a user must exert in order to interpret the state of the system and determine the extent to which expectations and intentions have been met. Feedback that is relevant, easily interpreted and clear helps to reduce this effort and minimise frustration, which the user may otherwise project onto a system through a negative perception of usability. Touchscreen devices have previously relied on visual feedback, again emphasising the visually-dominant nature of the technology. Removing visual feedback causes a significant increase in input error rates [9], as with physical keyboards [10] and mobile phone keypads [11].

Touchscreen design guidelines [12] state that visual feedback should be immediate in order to reassure the user that their input has registered; however no literature was found regarding the effects of latency in touchscreen visual feedback. The study of visual feedback latency in virtual environments has been more comprehensive (e.g. [13,14]), showing reductions in task performance and increased error rates; however the nature of the visual feedback in these cases is continuous rather than discrete and thus not directly analogous to touchscreen use.

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1.1. Multimodal interaction

The benefits of multimodal interaction have been widely studied. A meta-analysis of 43 studies which compared visual to visual–auditory and visual–tactile feedback [15] showed improvements in reaction times and performance scores over visual feedback alone, although no overall improvement was seen in error rates. This suggests that moving workload from the visual to the haptic channel can reduce cognitive demand.

Incorporating haptic feedback into a touchscreen interface enables users to receive feedback in both modalities employed in interaction. A number of studies have been conducted which specifically investigate the effects of haptic feedback in touchscreens: in mobile devices, haptic feedback was shown to reduce error rates for entry and selection tasks, reduce task completion time [16] and improve workload [17]. Lee and Spence [18] assessed the benefits of bimodal and trimodal feedback in a high-workload scenario, using a driving-themed reaction time test and touchscreen-based telephone secondary tasks. Their results showed that time taken to react to the driving task and to complete the telephone tasks were both reduced when trimodal (visual, haptic and auditory) feedback was enabled, with a corresponding reduction in subjective workload.

Further subjective benefits were seen in other studies: Lee et al. [19] investigated the effect of multimodal feedback on performance in older adults, using a bench-top emulation of a mobile phone interface. Results showed that while visual–auditory and visual–auditory–haptic feedback showed improvements in task performance, only trimodal feedback yielded reductions in perceived task difficulty. In the context of automotive use, Serafin et al. [20] showed that users preferred bimodal (visual plus auditory or haptic) feedback over visual alone, with tri-modal feedback strongly most preferred. Pitts et al. [21] conducted trials based on automotive touchscreen use cases scenarios in a simulated driving environment, with combinations of visual, auditory and haptic feedback. Again, tri-modal feedback was strongly most preferred by users, showing improvements in subjective difficulty rating, user confidence and driving task interference over visual feedback alone.

1.2. The nature of touchscreen interaction

Acting as both display and input, and in the absence of auditory feedback, touchscreen interaction employs the visual and haptic senses. For a menu-based interaction, the user employs target search visual scanning behaviour in order to locate the required target, the number of glances required depending upon interface design, expected target location and user experience [22]. Haptic interaction provides the input to the touchscreen and occurs once the target has been located, with feedback stimuli produced at this point. Further visual interaction may occur post-input through confirmation glances, employing supervisory control visual scanning behaviour. For the purposes of this study we define glances made prior to successful activation of the target as primary glances and those made after activation as secondary glances.

The nature of the haptic element of interaction is dependent on the feedback stimulus. Unlike vision, the haptic system makes use of both sensory (tactile) information and the movements required to obtain it [23]. Jones and Lederman [24] differentiate between passive tactile sensing, whereby the stationary hand is stimulated by either a stationary or moving object; and active haptic sensing, where the hand acts upon an otherwise static object. For a simple button-press action, haptic interaction with a touch screen at the point of contact may be considered passive, as there are no surface textures or features from which to discern information. Even with the introduction of vibrotactile feedback upon activation the interaction may still be considered passive; Jones and Lederman

again draw the distinction between a tactile display and a *haptic* display which incorporates kinaesthetic feedback. However, transient feedback stimuli, designed to replicate mechanical switch-feel characteristics and invoke the paradigm of switch use, may induce a sensation of longitudinal displacement [25]; the user ‘feels’ that the screen moves inwards when pressed. In this sense, one may argue that the perception of kinaesthesia is sufficient for the interaction to be considered active and hence the description of the display as ‘haptic’ is appropriate. The issue of the lack of tactile cues to assist in target identification and location still persists, however.

1.3. Study outline

While the literature discussed above offers evidence for the benefits of multimodal interaction over visual alone, the nature of the visual–haptic interaction and how it relates to real-world applications is unclear. A study was therefore conducted to investigate the interaction of visual and haptic feedback in an automotive touchscreen use case, using a simulated driving environment. Objective measures of visual behaviour, task and driving performance were employed, along with a range of subjective measures of user response. A repeated-measures experiment design was used in which participants performed touchscreen trials in the presence and absence of both visual and haptic feedback, while engaged in a driving task. In addition, an intermediate visual feedback condition was assessed whereby the visual stimulus is delayed; this was included both to investigate the extent to which the visual element dominates touchscreen interaction and to represent feedback latency issues found in some production vehicle interfaces.

2. Research questions and hypotheses

The study aims to address the following research questions:

- Does the introduction of haptic feedback affect objective driving performance?
- Does haptic feedback affect the visual workload demand of touchscreen use?
- Does haptic feedback remove the requirement for visual feedback?
- Does the presence of haptic feedback affect objective touchscreen task performance?
- Do users express a subjective preference for touchscreen haptic feedback?

Given the visual nature of touchscreen interaction, the hypothesis is formed that degradation of visual feedback will increase the visual workload, as measured by glance time and count. This increase in visual workload is expected to have a negative effect on driving performance. With the performance benefits seen for multimodal feedback described in Section 1.1, the introduction of haptic feedback is expected to reduce visual workload by moving a proportion of this workload to the haptic channel; however as feedback is produced at the point of haptic interaction this change in workload is expected to occur in secondary visual behaviour, as defined in Section 1.2. Finally, improvements in subjective workload, confidence and liking are expected with the introduction of haptic feedback.

3. Methodology

3.1. Participants

A total of 36 subjects (18 male) were recruited. Ages ranged from 20 to 65 years, (mean = 38.1 years, SD = 12.6 years) and all participants had at least 1 year experience of driving in the UK.

Subjects were recruited primarily from an existing list of participants, therefore most had some experience of simulator-based evaluation. In addition, two-thirds had experience of using touchscreen devices in their own cars, the majority of these being portable navigation devices (e.g. TomTom).

3.2. Equipment

The study was conducted using the driving simulator at the Human Factors Research Group, University of Nottingham.¹ This is a fixed-base system utilising a Honda Civic cabin and a 270° curved projection screen with three projectors providing full peripheral coverage. Separate LCD monitors display feeds for the rear- and side-view mirrors. The cabin is equipped with four miniature video cameras which capture driver behaviour. The simulation environment was generated using STISIM Drive software² which also recorded driver performance data.

The scenario employed for this study consisted of a simple motorway driving task. Users were required to follow a lead vehicle travelling at a constant 68 mph at what they perceived to be a 'safe' distance. The lead and driven vehicles were situated in the inside lane of motorway with three lanes in each direction separated by a central reservation. Traffic was included in both directions, with traffic vehicles travelling more quickly than the lead vehicle in lanes 2 and 3 while maintaining lateral lane position. Subjects were therefore not required to negotiate other vehicles and no safety-critical events such as sudden braking from the lead vehicle were included. This scenario replicates the conditions under which a driver is most likely to divert their attention to in-car tasks [26].

The touchscreen interface consisted of a Touchsense 8.4" demonstrator unit from Immersion Corporation,³ a resistive touchscreen with integrated haptic feedback actuators and controller. This was securely mounted into the centre console of the vehicle, providing a close approximation of the ergonomic conditions of a factory-installed system and serving as both the input and display devices for haptic and visual stimuli generated during the touchscreen tasks. Subjects were required to perform a series of touchscreen use tasks while engaged in the driving task, described below (see Fig. 1).

3.3. Evaluation interface

Previous work indicates that realistic use case tasks impose significant workload for non-expert users and that considerable training is required prior to evaluation [21]. The touchscreen task used for this study was an abstracted 'search and select' task whereby the participant is required to find and press a target button from a cluster displayed on the touchscreen. The interface was based on the design used by Irune and Burnett [27] to evaluate the effect of button size and cluster size on total glance time. The interface was programmed in Adobe Flash CS3 and Actionscript 3.0.

For each trial in the study, subjects were required to locate and press the button labelled "AC" from an array of buttons displayed on the touchscreen, placing the task in the context of operating the climate control system in a real vehicle. The other buttons in the array featured randomly-assigned labels which took the form "A_" or "_C", to create a degree of visual similarity to the target; no labels were duplicated within each task (Fig. 2a). A successful press of the target reset the screen ready for the next task. A 3 × 3 button array was chosen, corresponding to an expected total glance time of 2–3.5 s per task [27].



Fig. 1. Simulator cabin and touchscreen installation.



Fig. 2. Evaluation interface: (a) at start of task and (b) showing visual feedback.

Feedback was produced when a button was pressed. Visual feedback was presented as a change of button colour from black to white which lasted for a pre-determined period of 150 ms (Fig. 2b). In the delayed visual feedback condition, this occurred 1000 ms after the button press; responses within 250 ms may be considered 'immediate' [28], hence a 1000 ms delay should be clearly perceptible. In order to minimise variability, one haptic feedback effect was used throughout the trial, selected on the basis of preliminary user trials. This was a simple 'click' effect, analogous to the feel of a mechanical pushbutton, produced on press and release of the button. Users' affective response to touchscreen haptic feedback and task completion performance has been shown to be negatively affected by latency in haptic feedback [29,30], hence the haptic stimulus was always presented without induced delay.

3.4. Experiment design

The experiment utilised a 3 × 2 repeated measures (within-subjects) design, with three levels of visual feedback: 'immediate', 'delayed', 'none'; and two levels of haptic feedback: 'visual only', 'visual + haptic'. The experiment design was counterbalanced for gender and haptic feedback order. Participants completed two drives, with and without haptic feedback, each featuring 27 touchscreen tasks encompassing all combinations of target position and visual feedback condition. Each visual feedback level was therefore experienced nine times and each target position three times. Order of presentation was randomised for each drive.

The points at which the individual tasks occurred along the roadway were programmed into the simulator scenario and signalled by an audible tone. The task was then manually initiated by the experimenter, activating the touchscreen interface. Tasks

¹ University of Nottingham Human Factors Research Group: <http://www.virart.nottingham.ac.uk/>.

² STISIM Drive: <http://www.stisimdrive.com/>.

³ Immersion Corporation: <http://www.immersion.com/>.

were nominally spaced at 22.5 s intervals, plus or minus a randomly-assigned value of up to 5 s, giving a minimum spacing of 17.5 s and a maximum of 27.5 s. Two versions of the scenario were used to avoid direct replication of task spacing. Additional task events were included in the scenario to allow for subjects failing to complete a task within the prescribed period, with each experimental drive lasting approximately 11 min.

In order to maintain the focus of the experiment on the feedback stimulus itself, the behaviour of the button array displayed on the screen was standardised across all visual feedback conditions. A successful press of the target button initiated a 2000 ms period within which the visual feedback stimulus would be displayed according to the experimental condition. At the end of the 2000 ms period the screen reset in preparation for the next task.

The touchscreen haptic actuators produce sound when operated that is clearly audible in bench-top operation. If uncontrolled, an auditory–haptic cross-modal effect may occur that would serve to increase the perceived amplitude of the haptic stimulus. For this study, it was not appropriate to use headphones to provide isolation due to the immersive nature of the simulator; the level of the simulated road and engine noise within the cabin was therefore adjusted such as to subjectively mask the sound emitted by the haptic actuators and negate any cross-modal influence.

3.5. Training procedure

A three-part training procedure was employed. Firstly, participants completed a consent form and check-in questionnaire to provide details about their age, driving experience and touchscreen device usage. A pre-trial simulator sickness checklist was also completed, both to provide a baseline for the subject's condition and to make the subject aware of the possible symptoms which may be induced during interaction with the simulated environment. A second copy of this checklist was completed at the end of the study.

The second part of the training procedure was conducted inside the simulator. The participant was introduced to the touchscreen setup and was provided with a screen featuring two buttons, one with and one without haptic feedback enabled. All participants confirmed that they could perceive the haptic feedback at this stage. As the majority of participants were not from technical backgrounds, the term 'touch feedback' was used to refer to haptic feedback in the training and subsequent questionnaires.

The participant was then introduced to the evaluation interface and instructed on the touchscreen task as described in Section 3.3. This was followed by the completion of a series of practice tasks by the participant, featuring the different visual feedback conditions. While no strict performance criteria were set for the practice tasks, the experimenter ensured that participants were confident with the task prior to commencing the experimental drives. The programmed variations in visual feedback behaviour were not explicitly defined by the experimenter; rather participants were told that they 'might notice some differences' in the response of the buttons.

The final part of the training procedure involved a practice drive of approximately 6 min duration. This was conducted with the experimenter in the car to allow questions to be answered and no touchscreen tasks were conducted during this drive. Given the basic nature of the driving task, confirmation of satisfactory performance of the touchscreen task in a static condition and in order to minimise participants' exposure to the virtual environment, a dual-task practice drive was not conducted.

3.6. Data sources

Both objective and subjective data were collected during the study. Video recording of driver behaviour allowed observation of glance count and duration, as defined by ISO 15007-1:2002 [31].

Driving performance was measured through standard deviation of lane position (SDLP), mean speed, standard deviation of speed and time headway, while touchscreen task performance was measured through task completion time.

Subjective data were collected via a questionnaire administered after each drive, with an additional follow-up questionnaire at the end of the study. Participants were asked to rate their perception of hedonic rating (how much they liked using the touchscreen), confidence in button press, touchscreen task difficulty, interference with the driving task and, for the haptic-enabled case, the strength of the haptic feedback stimulus and the realism of the feedback (how much the feedback felt like a real switch). Hedonic rating used a 9-point hedonic rating scale with semantic anchors at each point [32], while remaining measures used 9-point scales with end and mid-point anchors. The follow-up questionnaire included a forced-choice most/least preferred selection, along with 5-point Likert scale ratings of the statements "Touch feedback makes the touchscreen easier to use" and "Touch feedback makes the screen more pleasurable to use".

3.7. Analysis

Objective data from 35 of the 36 subjects were analysed – one data set was lost due to technical issues. Visual events were identified using video editing software to manually code actions of interest, producing an event timeline which was exported for calculation of visual behaviour metrics. This was then synchronised with driving performance and task completion data to generate the metrics defined above.

Means values for each dependent variable were calculated, taking the average across the nine repeat measurements for each visual feedback state. Differences between group means were analysed using two-way within-subjects ANOVA, with visual and haptic feedback states as factors. Significance is determined at $\alpha = 0.05$ (two-tailed) for all tests.

4. Results

4.1. Effect of haptic feedback on glance duration

As defined in Section 1.2, visual interaction with a touchscreen interface consists of primary and secondary glances. The majority of the visual workload occurs in the primary phase and one would expect to see longer glance times compared to secondary glances.

A comparison of mean primary and secondary mean glance duration is shown in Fig. 3. Note that confidence limits for glance time are too small to illustrate on the figure and error bars are therefore omitted. Student's *t*-test shows no significant difference in mean primary glance time between the 'visual only' and the 'visual + haptic' states ($t(3865) = 0.50$; NS). Similarly, there was no significant difference found for secondary glance time ($t(784) = 0.57$; NS). Primary glances have approximately twice the duration of secondary glances on average.

4.2. Visual workload during task completion

Fig. 4 shows the mean primary glance time per task as a proportion of task completion time, i.e. the proportion of time that the participant spends looking at the touchscreen in order to successfully locate and press the target button. Over 70% of the task completion time is spent looking at the touchscreen, confirming the visual nature of touchscreen interaction. Again, confidence intervals for primary glance time are negligible and hence omitted from the figure. Analyses of mean primary glance and task completion time are made in Sections 4.4 and 4.6 respectively.

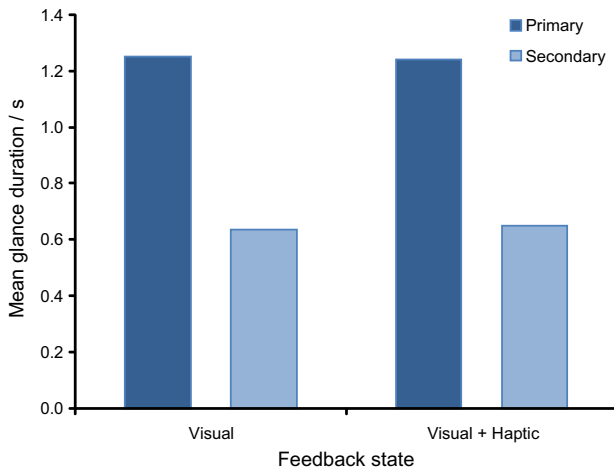


Fig. 3. Effect of haptic feedback on mean glance duration.

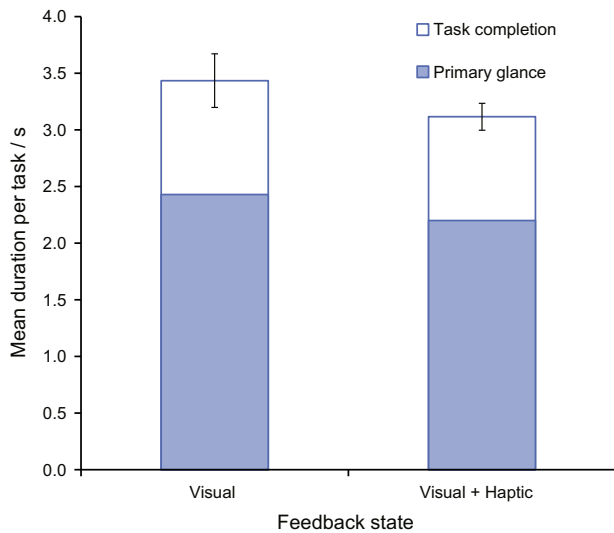


Fig. 4. Primary glance time as a proportion of task completion time. Error bars indicate 95% confidence interval for task completion time.

4.3. Total glance time

By taking the sum of all glance durations for each task, we obtain the total glance time (TGT) per task. The distribution for this measure is shown in Fig. 5 with extreme outliers removed; these were identified based on the inter-quartile range method described by Tukey [33] and defined as occurring outside the range

$$Q_2 \pm 3(Q_3 - Q_1)$$

where Q_2 is the median of the data set and Q_1 and Q_3 the 25th and 75th percentiles. Twenty data points were removed, corresponding to 1.06% of the data. Outlying data points occur when the participant fails to successfully complete the touchscreen task and reverts their attention to the forward roadway, requiring a second attempt to complete the task.

The mean TGT per task is 2.54 s while the median is 2.36 s, indicating positive skew of the distribution. The expected TGT range for this array size given by Irune and Burnett [27] of 2–3.5 s encompasses 50% of the data, from the 35th to 85th percentile.

Two-way within-subjects ANOVA analysis of mean TGT per task (Fig. 6) indicated a significant interaction between visual and haptic feedback ($F(2,68) = 6.177, p < 0.005$). Mean TGT was reduced when

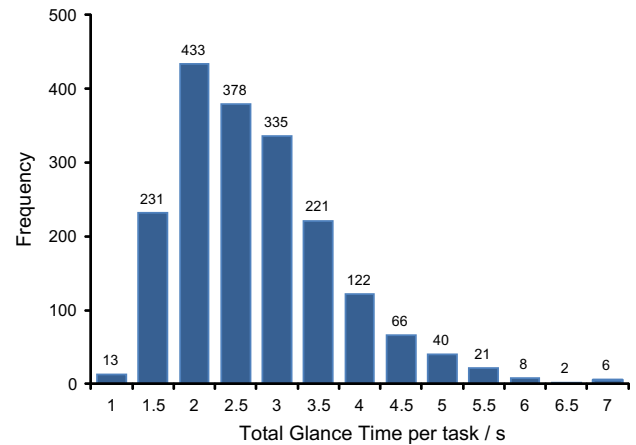


Fig. 5. Distribution of total glance time per task with outliers removed, for all experimental conditions ($n = 1870$).

haptic feedback was present; simple main effects analysis showed a statistically significant reduction from 2.76 s to 2.32 s for the 'delayed' visual feedback condition ($F(1,34) = 23.514, p < 0.001$), and from 2.96 s to 2.40 s for the 'none' condition ($F(1,34) = 40.329, p < 0.001$).

The main effect for visual feedback was significant when haptic feedback was not enabled ($F(2,68) = 12.050, p < 0.001$). A post-hoc Tukey HSD test for the 'visual only' state, corrected for family-wise type I error as per Cicchetti [34], indicated that the means for the 'immediate' and 'none' conditions were significantly different ($q(5,68) = 3.99, W = 0.25$; means = 2.53 s and 2.96 s respectively).

4.4. Primary glance behaviour

Further two-way ANOVA analyses were conducted on the primary glance data. Fig. 7 shows the mean primary glance time per task.

The main effect for visual feedback was not significant ($F(2,68) = 0.018, NS$), while interaction approached significance ($F(2,68) = 2.505, p = 0.089$). Mean primary glance time per task was reduced with the addition of haptic feedback; simple main effects analysis indicated a statistically significant reduction from 2.48 s to 2.15 s ($F(1,34) = 23.824, p < 0.001$) for the 'delayed' visual feedback condition, and from 2.45 s to 2.18 s for the 'none' condition ($F(1,34) = 15.893, p < 0.001$).

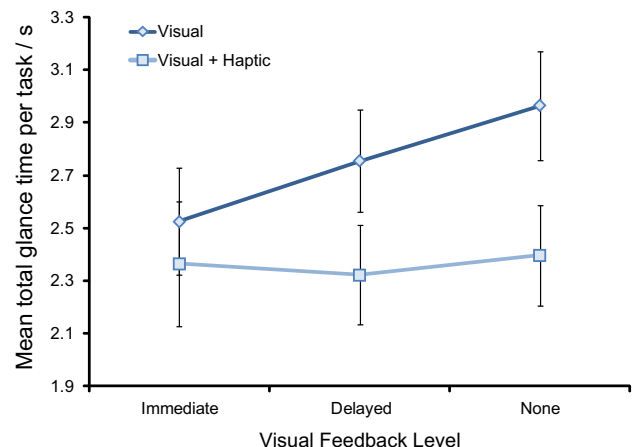


Fig. 6. Mean total glance time per task. Error bars indicate 95% confidence interval.

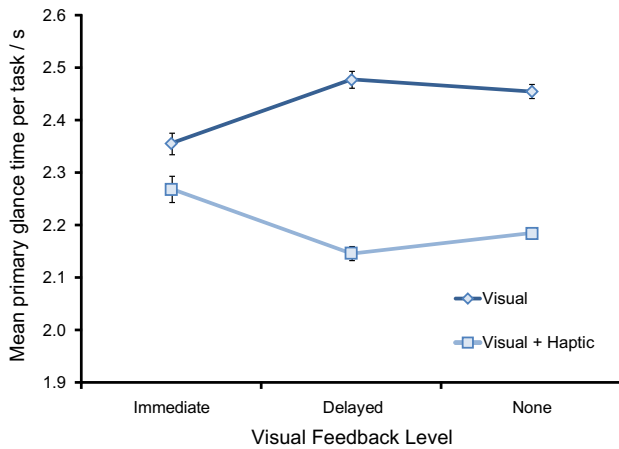


Fig. 7. Mean primary glance time per task. Error bars indicate 95% confidence interval.

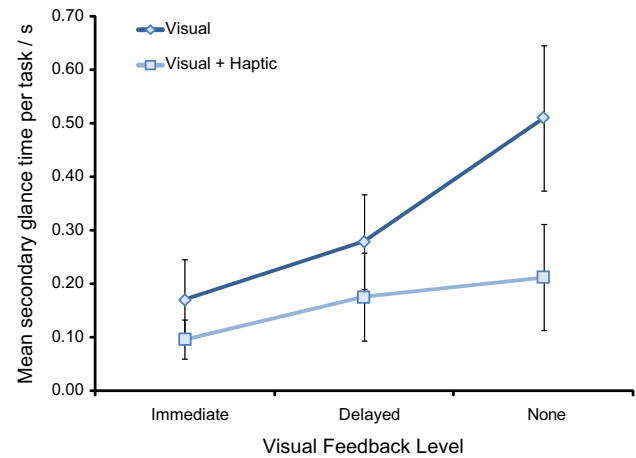


Fig. 9. Mean secondary glance time per task. Error bars indicate 95% confidence interval.

Given that the mean duration of primary glances did not change with the presence of haptic feedback (see Section 4.1), the number of glances made must differ between the test conditions. Fig. 8 show the glance count distributions for the 'visual only' and 'visual + haptic' states, with outliers removed.

Forty-six percent of tasks required only one primary glance with visual + haptic feedback, compared with 36% for visual feedback alone. A two-way ANOVA was also performed on primary glance count: only the main effect for haptic feedback was significant ($F(1,34) = 15.639$, $p < 0.001$) showing a reduction in overall mean from 1.94 to 1.77, a change of -8.8% in the number of primary glances made.

4.5. Secondary glance behaviour

Hypotheses formed prior to the study predicted an effect of haptic feedback on secondary visual behaviour. Fig. 9 shows the results for a two-way ANOVA on mean secondary glance time per task.

Interaction between visual and haptic feedback was shown to be significant ($F(1,34) = 13.895$, $p < 0.001$). Mean secondary glance time per task was reduced when haptic feedback was enabled; subsequent simple main effects analysis showed that the effect is statistically significant for the 'none' condition ($F(1,34) = 13.895$, $p < 0.001$), reducing the mean secondary glance time per task from 0.51 to 0.21 s.

The main effect for visual feedback was also found to be significant when haptic feedback was not enabled ($F(2,68) = 30.980$, $p < 0.001$). A post-hoc Tukey's HSD test (again corrected for

family-wise type I error) indicated significant differences between 'none' condition (mean = 0.51 s) and the 'immediate' and 'delayed' conditions (means = 0.17 s and 0.28 s respectively).

Again, as no difference was seen in the mean duration of individual secondary glances (see Section 4.1), the number of secondary glances must be affected by the inclusion of haptic feedback. The distributions of secondary glance count are shown in Fig. 10.

These data showed good fit to a Poisson distribution with expected number of glances = total number of glances/number of tasks ($\chi^2(2) = 2.73$, $p = 0.26$; $\chi^2(2) = 1.44$, $p = 0.49$ respectively). For both the 'visual only' and 'visual + haptic' states, the most likely outcome is for the user to make no secondary glances. The probability of making one or more secondary glances was reduced from 0.39 to 0.22 when haptic feedback is enabled, a change of -44% . Two-way ANOVA analysis of secondary glance count indicated that interaction was again significant ($F(2,68) = 12.573$, $p < 0.001$); mean secondary glance count per task was reduced when haptic feedback was enabled, reaching statistical significance for the 'none' visual feedback condition where glance count is reduced from 0.73 to 0.30, a change of -58% .

Main effect for visual feedback was significant when haptic feedback was not enabled ($F(2,68) = 26.701$, $p < 0.001$); Post-hoc Tukey's HSD test showed significant differences between all means for the 'visual only' state (Table 1).

4.6. Task completion time

Task completion time is defined as the time taken from the initiation of the touchscreen task to the successful location and press of

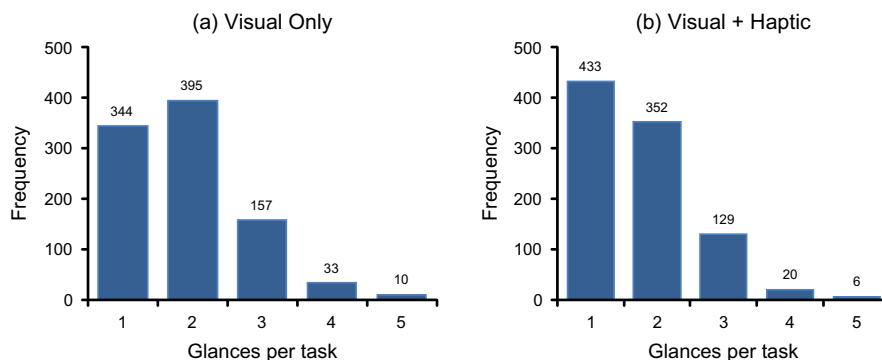


Fig. 8. Distributions of primary glance count shown with outliers removed for: (a) 'visual only' ($n = 939$) and (b) 'visual + haptic' ($n = 940$) states.

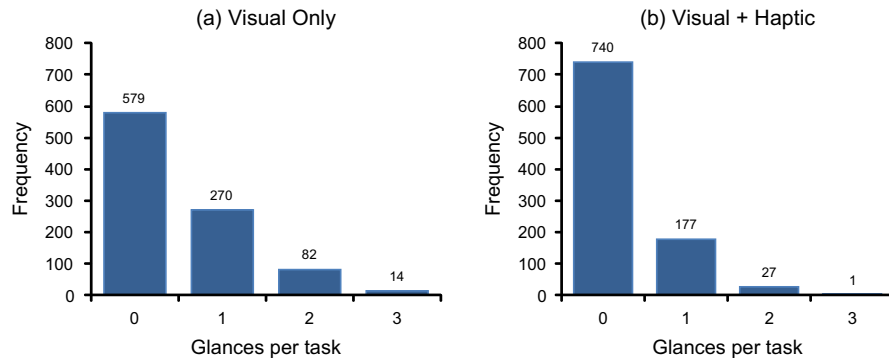


Fig. 10. Distributions of secondary glance count for: (a) 'visual only' and (b) 'visual + haptic' states ($n = 945$).

the target button. Mean task completion time for each condition was calculated and subjected to a two-way ANOVA. Only the main effect of haptic feedback was significant, showing a change in overall mean task completion time from 3.43 s to 3.09 s ($F(1, 34) = 6.938$, $p < 0.05$), a reduction of approximately 10%.

4.7. Driving performance

For the analysis of driving performance data were taken from the start of the task to 5 s after successful completion, thus allowing the driver time to regain their normal lane position and speed. The metrics derived were then pooled for each test condition and compared to a baseline for driving performance taken from a 3-min segment of the practice drive, where no touchscreen tasks were performed. The baseline condition was included as a fourth level in the visual feedback factor and the same values used for both haptic feedback states. A two-way ANOVA was subsequently performed to assess differences across conditions.

4.7.1. Standard deviation of lane position (SDLP)

Fig. 11 shows the results for standard deviation of lane position. Only the main effect for visual feedback was significant ($F(3, 102) = 13.901$, $p < 0.001$); post-hoc Tukey's HSD confirms that differences exist only between the baseline condition and other means ($q(6, 102) = 4.10$, $W = 0.140$). Therefore no effect on driving performance was attributable to changes in either haptic or visual feedback, but the introduction of the touchscreen tasks caused an increase in overall mean SDLP from 18.9 cm to 24.3 cm, an increase of 28%.

4.7.2. Speed variation

Analysis indicated that participants did not vary their speed relative to a baseline measure as a coping strategy while completing touchscreen tasks ($F(3, 102) = 0.944$, NS) and that no variation was observed due haptic feedback ($F(1, 34) = 0.001$, NS). Similarly, standard deviation of speed did not vary from the baseline measure ($F(3, 102) = 1.087$, NS) and was not affected by haptic feedback ($F(1, 34) = 0.154$, NS).

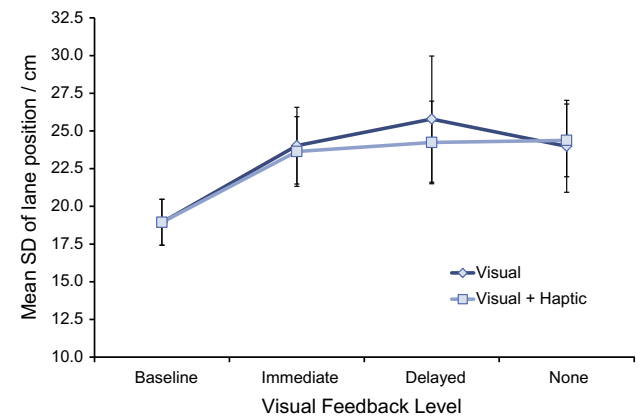


Fig. 11. Mean standard deviation of lane position. Error bars indicate 95% confidence interval.

4.7.3. Headway Variation

Mean time headway during completion of the touchscreen tasks was also shown not to vary significantly from the baseline measure ($F(3, 102) = 1.388$, NS); furthermore variation due to haptic feedback was not significant ($F(1, 34) = 1.424$, NS). Again, this indicates that participants did not increase their distance to the lead vehicle to cope with the dual-task demands.

4.8. Order analysis

To minimise bias, the presentation order of haptic feedback state was counterbalanced with equal numbers of participants experiencing each haptic feedback condition first. By repeating the two-way ANOVA analyses with order as a factor, it was possible to confirm that there were no significant effects due to haptic feedback presentation order for glance time (primary: $F(1, 34) = 0.130$, NS; secondary: $F(1, 34) = 0.041$, NS), glance count (primary: $F(1, 34) = 1.998$, NS; secondary: $F(1, 34) = 0.002$, NS) or task completion time ($F(1, 34) = 0.014$, NS).

Table 1
Post-Hoc Tukey's HSD table for secondary glance count – 'Visual only' state.

Visual feedback condition	Mean glance count per task	Difference between means		
		Immediate	Delayed	None
Immediate	0.302	–	0.175	0.432
Delayed	0.476		–	0.257
None	0.733			–

$q(5, 68) = 3.99$, $W = 0.168$.

Values in bold indicate significant difference.

4.9. Subjective ratings

Rating scale data from the post-trial questionnaires were analysed for significance using the non-parametric paired Wilcoxon signed-ranks test. Note that data for all 36 subjects were available for these measures. A summary of the subjective rating scale anchors used and mean scores is given in Table 2.

Hedonic rating showed a significant improvement of +1.3 rating points with the addition of haptic feedback ($W = 27$, $p < 0.001$), while confidence rating saw a larger improvement of +2.1 points ($W = 27$; $p < 0.001$). Small improvements in touchscreen task difficulty and driving interference were observed (+0.7, $W = 47.5$; $p < 0.01$ and +0.8, $W = 53.5$; $p < 0.01$ respectively). Note that in both cases the mean rating indicates a 'less than moderate' level of task difficulty/driving interference.

Users found that the haptic feedback stimulus strength was 'just right' (mean = 5.1) but did not accurately replicate the feel of a mechanical switch (mean = 3.9).

The additional subjective measures included in the follow-up questionnaire provided further evidence of user acceptance of the haptic touchscreen technology. For the forced-choice 'Most preferred' question, 32 of the 36 participants (89%) chose visual + haptic feedback as most preferred. Furthermore, subjects 'agreed' overall that haptic feedback made the touchscreen both easier and more pleasurable to use (Likert scale means of 4.2 and 4.1 respectively, see Fig. 12). Subjects agreed less strongly (mean = 3.7) that touchscreen haptic feedback should feel like mechanical push switches.

Finally, comparison between the pre- and post-study simulator sickness questionnaire responses indicated that no participants suffered detrimental effects from exposure to the simulated driving environment. Efforts made to screen participants and minimise exposure time are likely to have been beneficial in this regard.

5. Discussion

Results clearly showed that driver behaviour is affected both by the presence of a secondary task and by the types of feedback presented to them. The touchscreen interaction task used in this study was highly visual in nature, as indicated by the driver directing their gaze at the screen for over 70% of task completion time. When haptic feedback was not activated, total glance time per task increased when visual feedback was delayed or absent; degradation in visual feedback lead to an increase in visual workload. However, introducing haptic feedback counteracted this effect, with performance showing no significant difference from the optimal visual feedback condition. Therefore, while haptic feedback did not have a significant benefit when visual feedback was immediate, the total glance time per task was reduced in cases where visual feedback was delayed or absent, by 19% in the latter case.

Hypotheses formed prior to the study predicted that the effect of haptic feedback would be observed in secondary visual behaviour, i.e. after successful location and selection of the target. In fact, reductions in glance time with haptic feedback were found for both

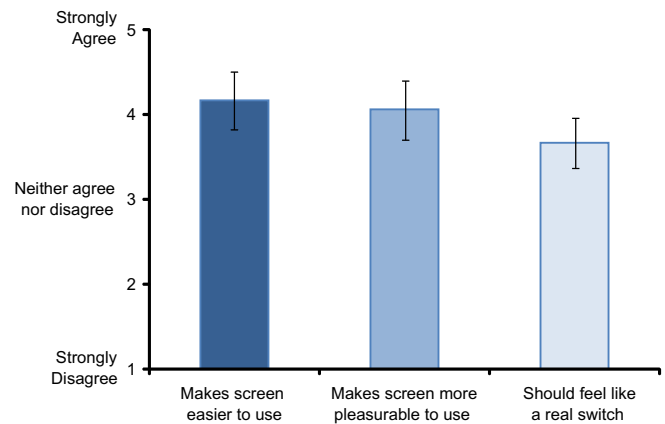


Fig. 12. User opinions of touchscreen haptic feedback, rated on a 5-point likert scale. Error bars indicate 95% confidence interval.

primary and secondary behaviour. Users were also able to complete more tasks with a single primary glance when haptic feedback was enabled, 46% compared to 36%. A corresponding reduction in task completion time was observed; this effect was equal for all visual feedback conditions.

The effect of delaying or removing visual feedback was to increase visual workload; this was observed as an increase in secondary glance time, as expected. Again, haptic feedback was found to counteract these effects, with no significant difference seen across visual feedback levels when haptic feedback was enabled. While the majority of tasks required no secondary glances, the probability of making a secondary glance was reduced when haptic feedback was enabled; this effect was statistically significant when visual feedback was absent, with secondary glance count reduced by 58%.

Visual workload is an important factor in the safety of an in-vehicle system. The model proposed by Wierwille [35] states that number of crashes due to visual distraction is directly proportional to glance count, and to mean glance duration to the power of 1.5. The reductions in glance time and count observed when haptic feedback was enabled will therefore have benefits in terms of safety. Perceived task difficulty and driving interference were rated 'moderate' or less throughout the study, indicating that the overall cognitive workload imposed by the abstracted touchscreen task and simple driving scenario were relatively low. Workload requirements, both cognitive and visual, might be expected to be significantly higher in a real-world use case, where significantly more glances and button presses may be required to complete a particular task. In such situations the premium placed upon visual attention may increase the benefits of supplying information via the haptic channel.

The expected effects of increased visual workload from a secondary task include degradation of lateral position control (i.e. increased deviation) and reduced speed [36]. Results from the study showed that standard deviation of lane position (SDLP) did

Table 2
Summary of subjective rating scale anchors and mean rating scores.

Measure	Scale anchors (score)			Mean rating		p
	Lower (1)	Midpoint (5)	Upper (9)	Visual	Visual + haptic	
Hedonic rating	Dislike extremely	Neither like nor dislike	Like extremely	5.3	6.6	<0.001
Confidence	Not at all confident	Moderately confident	Extremely confident	5.1	7.2	<0.001
Task difficulty	Extremely difficult	Moderately difficult	Not at all difficult	5.4	6.3	<0.01
Driving interference	Extreme interference	Moderate interference	No interference	5.0	5.8	<0.01
Feedback strength	Too weak	Just right	Too strong	–	5.1	–
Feedback realism	Not at all like a real switch	Somewhat like a real switch	Identical to a real switch	–	3.9	–

indeed increase with the introduction of the secondary task, by approximately 28% relative to a baseline measure; this concurs with the findings of Rydstrom et al. [37] who observed similar results from a study involving haptically-enhanced in-car rotary controls. No changes were observed however in either speed or headway control.

Subjective responses indicate that participants expressed greater liking and were more confident in using the touchscreen with haptic feedback enabled (the latter being the stronger effect); a small but significant decrease in the perceived difficulty of the touchscreen task was also observed. As participants also 'agree' on average that haptic feedback makes the touchscreen both easier and more pleasurable to use, it is implied that haptic feedback improved user experience and increased driver 'comfort' in terms of the combined performance of driving and touchscreen interaction tasks. This is important in the wider context of the automotive market, as user controls which are perceived as 'difficult to use' can have a negative effect on the perception of quality of the vehicle as a whole, with a subsequent negative effect on sales [38].

Participants' subjective assessment of the degree of interference with the driving task showed a small but significant improvement with haptic feedback; participants felt that their driving was less impaired when, objectively, there was no difference. Again, this adds further evidence for an overall improvement in the user experience when haptic feedback is enabled. The exclusion of auditory stimuli from the study means that relative effects of bimodal and trimodal feedback cannot be determined, but allowed a focused experiment design which generated reliable data.

The use of a simulated driving environment and abstracted secondary task impose some limitations on this study. Users may moderate their visual workload and driving behaviour differently while driving a real vehicle and more involved in-vehicle tasks may have a disproportionately greater effect on driving performance. For example, using a navigation system can produce an increase of 80% in lateral deviation [39], compared to the 28% observed here. However, a recent study by Wang et al. [40] found that visual attention measures and task response times for technology interaction tasks conducted in a fixed-based, medium-fidelity driving simulator (similar to that used in this study) mapped closely to those from an instrumented vehicle on real roads, providing support for the validity of a simulator-based HMI evaluation approach.

6. Conclusions

This study was designed to investigate the interaction between visual and haptic feedback in a touchscreen interface. Changes observed in glance behaviour, task completion and driving performance illustrate the requirement for confirmatory feedback in this type of interaction. Delaying or removing visual feedback increased total glance time as expected; however, the introduction of haptic feedback counteracted this change with no significant difference observed from the optimal visual feedback condition. As such, haptic feedback was able to compensate for the information loss which occurred when visual feedback was degraded. The effects observed here may be more prominent in real-world use cases, where workload requirements from both driving and technology interaction are likely to be significantly higher.

Reductions in glance time and count with haptic feedback were observed in both primary and secondary measures, which will have a corresponding benefit in terms of safety as in-vehicle visual workload is correlated to accident risk. However it is important to note that the effects of secondary tasks on objective driver performance may not be mitigated by this type of haptic feedback. Touchscreen interaction still requires drivers to take their eyes off the road and their hand off the steering wheel in order to locate

and perform their required function; again, the difference between baseline and in-task measures of driving performance seen in this study may be greater in a real-world scenario.

There were also additional effects attributable to haptic feedback, characterised by improved user experience and confidence in the touchscreen interface. While subjective measures of user experience may be susceptible to some degree of 'halo' effect from the experience of a novel technology, when viewed in conjunction with corresponding objective benefits there is compelling evidence for a tangible effect on driver comfort. Vehicle manufacturers should be aware of this when considering future HMI applications, as usability and affective response to in-vehicle technology play a major part in the perception of quality of the vehicle as a whole.

This work could be extended by the investigation of visual-haptic feedback touchscreen interaction in scenarios with varying primary and secondary task workloads. Performing further evaluations in an on-road study would provide validation of the simulator-derived results. Further analysis may be applied to existing data to examine for trends due to age, gender or technology experience.

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